

CM0081 Formal Languages and Automata

§ 3.1 Regular Expressions

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Preliminaries

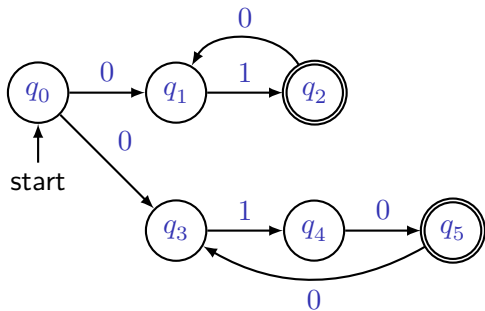
Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Introduction: Description of regular languages

$$(01)(01)^* + (010)(010)^*$$

Algebraic description



Machine-like description

Introduction: Regular Expressions

Features

- Algebraic description of regular languages

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- Declarative (“user-friendly”) way to express the strings that belong to the language

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Introduction: Regular Expressions

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Uses

- Search commands (e.g. `GREP`)
- Lexical-analyzer generators (e.g. `LEX` and `ALEX`)
- Domain specific languages (DSLs)

Operations on Languages

Notation

The power set of a set A is denoted $\mathcal{P}A$.

Operations on Languages

Definition

Let L , L_1 and L_2 be languages on an alphabet Σ .

(i) **Union** of languages:

$$\begin{aligned} \cup : \mathcal{P}\Sigma^* \times \mathcal{P}\Sigma^* &\rightarrow \mathcal{P}\Sigma^* \\ L_1 \cup L_2 &:= \{x \mid x \in L_1 \text{ or } x \in L_2\}. \end{aligned}$$

(iii) **Powers** of a language:

$$\begin{aligned} (-)^{(-)} : \mathcal{P}\Sigma^* \times \mathbb{N} &\rightarrow \mathcal{P}\Sigma^* \\ L^0 &:= \{\varepsilon\}, \\ L^{n+1} &:= L \cdot L^n. \end{aligned}$$

(ii) **Concatenation** of languages:

$$\begin{aligned} \cdot : \mathcal{P}\Sigma^* \times \mathcal{P}\Sigma^* &\rightarrow \mathcal{P}\Sigma^* \\ L_1 \cdot L_2 &:= \{x \cdot y \mid x \in L_1 \text{ and } y \in L_2\}. \end{aligned}$$

(iv) **Kleene closure** of a language:

$$\begin{aligned} (-)^* : \mathcal{P}\Sigma^* &\rightarrow \mathcal{P}\Sigma^* \\ L^* &:= \bigcup_{n \geq 0} L^n. \end{aligned}$$

Operations on Languages

Examples

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- If $L = \{0^n \mid n \geq 1\}$, then $L^* = L \cup \{\epsilon\}$.
- If $L = \{0, 11\}$, then L^* consists of the empty word and those strings of 0's and 1's such that the 1's come in pairs.

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- If $L = \{0, 11\}$, then L^* consists of the empty word and those strings of 0's and 1's such that the 1's come in pairs.
- Powers on $\emptyset$

$$\emptyset^0 = \{\varepsilon\},$$

$$\emptyset^i = \emptyset, \quad \text{for } i \geq 1,$$

$$\emptyset^* = \{\varepsilon\}.$$

What the Regular Expressions Are

Definition

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 - (i) ε is a regex,
 - (ii) $\emptyset$ is regex and
 - (iii) If $a \in \Sigma$ then a is a regex.
- Inductive step
 - If E and F are regex's then
 - (i) $E + F$ is a regex,
 - (ii) $E \cdot F$ is a regex,
 - (iii) E^* is a regex and
 - (iv) (E) is a regex.

Precedence of Operators

Order of precedence and associative

Precedence from highest to lowest: $()$, $*$, $\cdot$ and $+$.

Associative: The operators $\cdot$ and $+$ are left-associative.

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Example

$$\begin{aligned}01^* + 1 &= (0(1^*)) + 1 \\ &\neq (01)^* + 1 \\ &\neq 0(1^* + 1)\end{aligned}$$

Languages Denoted by Regular Expressions

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- Inductive step

Let $L(E)$ and $L(F)$ be the languages denoted by the regular expressions E and F , then

$$L(E + F) := L(E) \cup L(F),$$

$$L(E \cdot F) := L(E) \cdot L(F),$$

$$L(E^*) := (L(E))^*,$$

$$L((E)) := L(E).$$

Languages Denoted by Regular Expressions

Example

E	$L(E)$
$a + b$	$L(\mathbf{a}) \cup L(\mathbf{b}) = \{a\} \cup \{b\} = \{a, b\}$

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a^*	$\{\varepsilon, a, aa, aaa, \dots\}$

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$a + b$	$L(a) \cup L(b) = \{a\} \cup \{b\} = \{a, b\}$
a^*	$\{\varepsilon, a, aa, aaa, \dots\}$
$(a + b)(a + b)$	$L(a + b) \cdot L(a + b) = \{a, b\} \cdot \{a, b\} = \{aa, ab, ba, bb\}$

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$a + (ab)^*$	$\{a, \varepsilon, ab, abab, ababab, \dots\}$

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$a + (ab)^*$	$\{a, \varepsilon, ab, abab, ababab, \dots\}$
$(0 + 1)^*01(0 + 1)^*$	$\{x01y \mid x, y \in \{0, 1\}^*\}$

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$a + (ab)^*$	$\{a, \varepsilon, ab, abab, ababab, \dots\}$
$(0 + 1)^*01(0 + 1)^*$	$\{x01y \mid x, y \in \{0, 1\}^*\}$
$a_i(a_1 + a_2 + \dots + a_n)^*$	$\{w \in \Sigma^* \mid w \text{ starts by } a_i\}$

Languages Denoted by Regular Expressions

Example

Write a regular expression for the language L defined by

$$L = \{ w \in \{0, 1\}^* \mid 0 \text{ and } 1 \text{ alternate in } w \}.$$

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Write a regular expression for the language L defined by

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Solution.

$$(01)^* + (10)^* + 0(10)^* + 1(01)^*$$

Languages Denoted by Regular Expressions

Example

Write a regular expression for the language L defined by

$$L = \{ w \in \{0, 1\}^* \mid 0 \text{ and } 1 \text{ alternate in } w \}.$$

Solution.

$$(01)^* + (10)^* + 0(10)^* + 1(01)^*$$

Other solution.

$$(\varepsilon + 1)(01)^*(\varepsilon + 0)$$

Languages Denoted by Regular Expressions

Example

The regular expression

$$(10 + 0)^*(\varepsilon + 1)$$

denotes the set of strings of 0's and 1's that have no two adjacent 1's.

Languages Denoted by Regular Expressions

Example

Write a regular expression for denoting the set of strings over $\Sigma = \{0, 1\}$ not ending in 01 .

Languages Denoted by Regular Expressions

Example

Write a regular expression for denoting the set of strings over $\Sigma = \{0, 1\}$ not ending in **01**.

Solution.

$$\varepsilon + \mathbf{0} + \mathbf{1} + (\mathbf{0} + \mathbf{1})^*(\mathbf{00} + \mathbf{10} + \mathbf{11})$$

Derivatives of Regular Expressions

Observation

The material on derivatives of regular expressions is from (Brzozowski 1964).

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Definition

Let $L \subseteq \Sigma^*$ be a language and $a \in \Sigma$ a symbol. We define the **derivative** of L by a , denoted by $\partial_a L$, by

$$\begin{aligned}\partial_a &: \mathcal{P}\Sigma^* \rightarrow \mathcal{P}\Sigma^* \\ \partial_a L &= \{ x \in \Sigma^* \mid ax \in L \}.\end{aligned}$$

Derivatives of Regular Expressions

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The material on derivatives of regular expressions is from (Brzozowski 1964).

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Example

$$\begin{aligned}\partial_a \{abab, abba\} &= \{bab, bba\}, \\ \partial_a L(ab^*) &= L(b^*), \\ \partial_b L(ab^*) &= \emptyset.\end{aligned}$$

Derivatives of Regular Expressions

Definition

Let E be a regular expression on Σ and let $a \in \Sigma$ be a symbol. We define recursively the **derivative** of E by a , denoted $\partial_a E$, by

$$\partial_a : \text{RegEx} \rightarrow \text{RegEx}$$

$$\partial_a \emptyset = \emptyset,$$

$$\partial_a \varepsilon = \emptyset,$$

$$\partial_a \mathbf{a} = \varepsilon,$$

$$\partial_a \mathbf{b} = \emptyset, \quad \text{for } a \neq b,$$

$$\partial_a (E + F) = \partial_a E + \partial_a F,$$

$$\partial_a (EF) = \begin{cases} (\partial_a E)F + \partial_a F, & \text{if } \varepsilon \in L(E), \\ (\partial_a E)F, & \text{otherwise,} \end{cases}$$

$$\partial_a (E^*) = (\partial_a E)E^*.$$

Derivatives of Regular Expressions

Definition

Let E be a regular expression on Σ and let $w \in \Sigma^*$ be a string. We define recursively the **derivative** of E by w , denoted $\partial_w E$, by

$$\begin{aligned}\partial_w &: \text{RegEx} \rightarrow \text{RegEx} \\ \partial_\varepsilon E &= E, \\ \partial_{ax} E &= \partial_a(\partial_x E).\end{aligned}$$

Derivatives of Regular Expressions

Theorem (Brzozowski (1964), Theorem 4.2)

Let E be a regular expression on Σ and let $w \in \Sigma^*$ be a string. Then

$$w \in L(E) \quad \Leftrightarrow \quad \varepsilon \in L(\partial_w E).$$

Libraries

Observation

Theoretical regular expressions $\neq$ practical regular expressions.

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Some programming languages with support to regular expressions

.NET, C, HASKELL, JAVA, MATHEMATICA, MATLAB and PERL.

Algorithms

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See the `HASKELL` implementation of some algorithms on regular expressions in the course homepage.

Applications

Some programs that use regular expressions

GREP: Print lines matching a pattern

AWK: Pattern scanning and processing language

SED: Stream editor for filtering and transforming text

ALEX, FLEX and LEX: Lexical-analyser generators

EMACS and VIM: Text editors

MYSQL and ORACLE: Databases

Applications

Reading

§ 3.3. Applications of Regular Expressions.

Reading

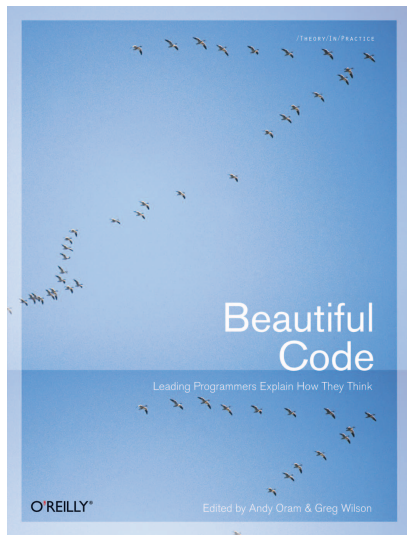
§ 3.3. Applications of Regular Expressions.

In the above section are defined:

$E^+ := EE^*$ (one or many times operator)

$E? := \varepsilon + E$ (zero or one time operator)

An Implementation: A Regular Expression Matcher



“Rob’s implementation itself is a superb example of beautiful code: compact, elegant, efficient, and useful. It’s one of the best examples of recursion that I have ever seen.”

Brian Kernighan, p. 3.

References



Janusz A. Brzozowski (1964). Derivates of Regular Expressions. Journal of the ACM 11.4, pp. 481–494. DOI: [10.1145/321239.321249](https://doi.org/10.1145/321239.321249) (cit. on pp. 35–37, 40).



John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).